

Chapter 9

OF MASS, CHARGE AND SPIN, THE BASIC ATTRIBUTES OF MATTER – THEIR PHYSICAL ORIGIN

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Abstract

The Standard Model of particle physics provides our inceptive understanding regarding the physical origin of mass, charge and spin – the basic attributes of matter. With an overview of the different aspects of these basic attributes, the present status of our understanding is assessed in this article.

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1. In lieu of an Introduction

According to the current wisdom, inanimate objects in our reality pronounce existence through the manifestation of (i) inertia against change of state or gravity, resulting from the curvature of space-time in the presence of matter; (ii) electrostatic or magnetic fields due to the presence of static or moving electric charge respectively and (iii) their ability to radiate energy (resulting from the acceleration or oscillation of electric charges). The most commonly acknowledged source of magnetic field, on the other hand, is a magnetic dipole arising from the orbital and most importantly ‘intrinsic spin’ motions of elementary charged particles at the microscopic level. Their ‘intrinsic spin’ also appear to dominate magnetism – which is usually referred to describe the response of material bodies to an applied magnetic field. Moreover, the spin of a particle has crucial role in the *statistical behaviour* of ensemble of identical particles. Thus, mass, charge and spin are regarded as the basic attributes of matter. With a brief review of the different aspects of these basic attributes, we shall try to assess in this article the present status of our understanding regarding their physical origin.

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Classical (Newtonian) mechanics introduces mass as a conserved and irreducible property of matter. Defining it as an isotropic scalar invariant¹, this formulation tacitly makes no distinction between the inertial and gravitational mass. Electrically charged matter, on the other hand, is influenced by, and produces electromagnetic fields. Charged particles whose charges have the same sign² (polarity) repel one another, and particles whose charges have different signs attract. Particle mass and charge are related to the ‘coupling constants’, representing the strength of gravitational and electromagnetic interactions respectively – the two of the four fundamental interactions of nature. Aside from gravity, all the macroscopically observable forces – such as friction and pressure are due to electromagnetic interaction. The other two (strong and weak nuclear) interactions are only seen among subatomic elementary particles and atomic nuclei.

Electric charge is a characteristic property of many subatomic particles – the charge of an antiparticle equals that of the corresponding particle, but with opposite sign. The electric charges of free-standing particles are integral multiples of the elementary charge e (magnitude of electronic charge); we say that electric charge is quantized. By convention, the charge of an electron is $-e$, while that of a positron or proton is $+e$.

Also, charge is a relativistic invariant. This means that the charge q of any particle will always remain q , no matter how fast it moves. This property has been experimentally verified by showing that the charge of one helium nucleus (two protons and two neutrons bound together in a nucleus and moving around at high speeds) is the same as two deuterium nuclei (one proton and one neutron bound together, but moving much more slowly than they would if they were in a helium nucleus).

Another very important aspect, expressed by the charge-current continuity equation that follows from Maxwell’s electromagnetic theory is the conservation of charge of an isolated system. In essence it implies that the total amount of charge in the universe is constant, but leaves open the important question of what that amount is. Most evidences indicate that the total charge in the universe is zero; that is, there are equal amount of positive and negative charges. On the other hand, a central idea in theoretical physics – a consequence of E. Noether’s theorem³ – asserts that each conservation law is associated with a symmetry of the problem. Also in this context comes H. Weyl’s seminal contribution arguing the association of charge conservation with the ‘gauge’ or scale invariance [3] of the electromagnetic field⁴.

In the classical framework, mass is a primary concept and the question, ‘What is the origin of mass?’ makes no sense at all! Without a proper understanding of the microscopic level, classical physics is also inadequate to address the question regarding the origin of the ‘electric charge’ and finally the *spin of an elementary particle* is essentially a nonclassical concept⁵. In Classical mechan-

¹Speculation based on Mach’s principle that inertia of a body, which originates in a kind of interaction with all other bodies in the universe, may be anisotropic is not vindicated by carefully designed experiments [1]. H.-J. Treder [2], on the other hand, observed that Mach’s idea does not necessarily imply anisotropy of inertial masses and “the Mach-Einstein doctrine is compatible with the isotropy of mass in each cosmos”.

²It is now a well accepted idea that there is only one kind of electric charge (the Franklin-Watson model), and only one variable is needed to describe electrical charge which can be represented as a point along a one-dimensional number line – the positive charge is represented by point to the right (of zero) on the number line, while negative charge is represented by point to the left.

³Noether presented this result shortly after her arrival in Göttingen in 1915 (published in 1918) and Einstein regarded this as a piece of ‘penetrating mathematical thinking’.

⁴In classical electromagnetism we have the simplest example of gauge invariance reflected in the arbitrariness of the scalar and vector potentials (ϕ and $\mathbf{A}$ respectively) in representing the observable electric and magnetic fields to the extent:

$$\phi' = \phi - \frac{\partial \chi}{\partial t}; \quad \mathbf{A}' = \mathbf{A} + \nabla \chi,$$

χ being any arbitrary scalar field. Such transformations leave all measurable physical quantities unchanged.

⁵To produce the necessary angular momentum, the electron’s hypothetical surface would have to be moving faster than the speed of light and violate the theory of relativity. Also, a g-factor (gyromagnetic ratio/factor) of 2 for the spin magnetic moment of an electron is twice what one would expect if the electron spin were treated

ics, the distinction between spin and orbital angular momentum is merely a matter of convenience as ‘classical spin’ can always be expressed as the sum total of the orbital angular momenta of the constituents of the system. Spin in quantum mechanics (QM), on the other hand, implies a nonorbital, fixed angular momentum which is independent of inertial property (mass) and angular velocity, i.e. a truly intrinsic physical property, akin to the *elementary particle’s* electric charge and mass.

Just like elementary particles, composite particles such as protons, neutrons and atomic nuclei etc. are also referred to as having a definite spin in QM. It is understood to imply the total angular momentum of the lowest-energy internal state of the composite particle subject to the same quantization condition.

The fine structure splitting observed in atomic spectra was one of the first experimental evidences for the introduction of the concept of electron spin⁶. While reviewing the high resolution experiment showing fine structure in sodium D-line, W. Pauli [7] in 1924-25 proposed a new ‘two valued degree of freedom’ for electrons and advanced the *exclusion principle* stating that no two electrons (fermions) can share the same quantum state at the same time. R. de L. Kronig subsequently suggested a physical interpretation of Pauli’s new ‘degree of freedom’ in terms of electron spin which was greeted with severe criticisms from both Pauli and W. Heisenberg. Being ultimately convinced with Pauli’s argument, Kronig then refrained from publishing his ideas. However, it was also recognized in the early years of QM that the anomalous Zeeman effect in atomic spectra (observed in an external magnetic field) cannot be explained with just three quantum numbers n , l , and m . Finally in 1926, intrepid S. A. Goudsmit and G. E. Uhlenbeck [8] interpreted the doublet structure in hydrogen spectrum with the introduction of spin angular momentum in addition to the usual angular momentum due to the orbital motion. The explanation of the fine structure splitting of atomic spectra in terms of the spin-orbit coupling received gradual acceptance with L. H. Thomas’ relativistic correction [9] removing the ‘factor of two’ discrepancy between the theory and experimental observation (for an excellent historical account see ref.[10]). The theory allows the electron to be magnetic with the appropriate magnetic moment and successfully explain the complicated Zeeman effect.

Spin obeys similar mathematical laws (commutation relations) as do quantized angular momenta although it can not be pictured as the rotation of an elementary particle around its own axis and may have half-integer quantum numbers. Apart from the Pauli exclusion principle electron spin is the key to the understanding of the periodic system of chemical elements. Electron spin plays an important role in magnetism, with applications for instance in computer memories and spin of a photon is associated with the polarization of light. Manipulation of electronic spins in semiconductor devices is the subject of the developing field of ‘spintronics’. The manipulation of nuclear spin by radio frequency waves (nuclear magnetic resonance) is important in chemical spectroscopy and medical imaging.

Revelations of some weird and counterintuitive quantum mechanical aspects of nature are now being realized in laboratory experiments regularly [6] and contributing to the tremendous technological growth. With the advent of Theory of relativity and Quantum mechanics, some basic issues became increasingly relevant and we are in a position to ask and probe questions regarding the origin of the basic attributes of matter.

as a classical current loop.

⁶The first direct observation of the electron’s spin angular momentum was achieved much earlier (1922) in the Stern-Gerlach (S-G) experiment. But it was initially hailed to provide the experimental confirmation of a twofold space quantization of orbital angular momentum. Even the postulation of spin in 1926 could not led to a reinterpretation of S-G splitting [4]. The attribution of this splitting to spin came only after the observation of R. G. J. Fraser [5] in 1927 that the ground state orbital angular momentum and associated magnetic moments of silver, sodium and hydrogen are zero. Equally important aspect is the fact that, this (S-G) experiment provides a direct empirical basis of the theoretical structure of quantum mechanics [6].

The two fundamental equations of Special theory of relativity (STR) are given by:

$$E^2 = m^2 c^4 + c^2 \mathbf{p}^2 \quad (1)$$

$$\text{and } \mathbf{p} = m\gamma \mathbf{v} \quad (2)$$

where, E and $\mathbf{p}$ represent energy and momentum respectively of a massive particle moving with velocity $\mathbf{v}$ of mass m (Newtonian), c is velocity of light and γ being the Lorentz factor ($1/\sqrt{1 - \mathbf{v}^2/c^2}$). The product $m\gamma$ is called the relativistic mass representing a velocity dependence of the moving mass, while m is termed as rest mass. Combining these two eqns. we get, $E = m\gamma c^2$ and for a particle at rest, $\mathbf{v} = 0 \implies \gamma = 1$, we write $E_0 = mc^2$. However, “the world’s most famous equation” is usually written as

$$E = mc^2,$$

and the context tells us which mass is implied. Together with the force equation given by:

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = m\gamma \frac{d\mathbf{v}}{dt} + \frac{m\gamma^3 \mathbf{v}}{c^2} \cdot \frac{d\mathbf{v}}{dt}, \quad (3)$$

eqns. (1) and (2) define the relativistic dynamics. It may be noted that this formulation is free from the erroneous concept of ‘longitudinal’ or ‘transverse’ mass of early years.

There are two equivalent ways of formulating special relativity (some historical details can be found in [11]). In the above Einstein’s original formulation of STR, two separate conservation principles of classical physics are fused in a combined mass-energy conservation principle. The concept of relativistic mass is used to show a correspondence with Newtonian mechanics in the limit of small velocities, but it is not essential. The alternative geometric formulation introduced afterwards by H. Minkowski is described in terms of space-time, energy-momentum four vectors, world lines and considers m as the invariant mass, without using the notion of relativistic mass at all. The Minkowski-space description is also useful in the case of General theory of relativity.

The usefulness (and even the validity) of the concept of relativistic mass has raised a stormy debate [12] for quite some time. Einstein himself wrote in a 1948 letter to L. Barnett “It is not good to introduce the concept of the mass $M = m/\sqrt{1 - (v/c)^2}$ of a body for which no clear definition can be given. It is better to introduce no other mass than ‘the rest mass’ m ”. However, as we will see following F. Wilczek [13], most of the rest mass of composite particles ultimately consists of the relativistic mass of the constituent elementary particles. The idea of invariant (rest) mass is perhaps best adapted in the case of elementary particles, which are ‘irreducible’.

Nineteen seventies saw the rapid emergence of a paradigm – The Standard Model (SM) of particle physics based on the theory of relativity and quantum mechanics. Physicists are now searching the origin of mass within the framework of Standard Model. In this model it is argued that matter acquires mass through what is termed as ‘Higgs mechanism’ and due to asymptotic freedom. On the other hand, alternative theories are proposed to explain the ‘origin of quantum mechanical spin’. However, the ‘origin of charge’ is relatively less discussed and far less understood.

2. Ingredients of the Basic Theory

Shaking off the early reservations following the resolution of spin-orbit coupling problem by Thomas, Pauli [14] ultimately formalized the theory of spin in 1927, by grafting ‘spin’ into the nonrelativistic Schrödinger equation. He pioneered the use of Pauli matrices:

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \text{and} \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (4)$$

as a representation of the spin operators ($\hat{S}_x = \sigma_1 \hbar/2$ etc.), and introduced a two-component spinor statefunction for spin- $\frac{1}{2}$ electrons. Pauli's theory of spin was nonrelativistic. In the very next year (1928), P. Dirac [15] came-up with his relativistic quantum equation of motion which removes the problem of negative probability density of O. Klein [16] and W. Gordon's [17] relativistic generalization of Schrödinger equation⁷ and provides natural description of particle spin.

Dirac was able to linearize the relativistic energy expression (1) in a quantum mechanical operator form using (4×4) matrices in the lowest order – the so-called Dirac matrices. The Lorentz invariant equation of motion involving first order space-time derivatives, using a four-component spinor field for the statefunction of electrons is given by:

$$(i\hbar\gamma_\mu \frac{\partial}{\partial x_\mu} - mc)\psi = 0; \quad \mu = 0, 1, 2, 3. \quad (5)$$

Where, the Dirac matrices are,

$$\gamma_0 = \begin{bmatrix} I_2 & 0 \\ 0 & -I_2 \end{bmatrix} \quad \text{and} \quad \gamma_i = \begin{bmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{bmatrix} \quad i = 1, 2, 3$$

— I_2 being the 2×2 identity matrix and σ_i are the Pauli matrices.

Dirac's theory of electron leads to Pauli equation in the nonrelativistic limit. The four-component solution consists of symmetrical positive and negative energy solutions each having two components – representing two possible intrinsic states of a particle. The proper interpretation of the negative energy solutions as representing antiparticles with positive energies is obtained from quantum field theory (QFT). The relativistic quantum theory of electron formulated by Dirac finds an effective combination with the theory of gauge invariance mentioned earlier and together provide the basis of QFT – one of the corner stone of modern physics.

As a qualitative concept, it is often handy to picture spin angular momentum classically (as a vector). For instance, quantum mechanical spin can exhibit phenomena analogous to classical gyroscopic effects. One can exert a kind of 'torque' on an electron by putting it in a magnetic field and the 'spin vector' undergoes precession, just like a classical gyroscope. However, a proper description of quantum mechanical spin is obtained by spinors, and not by vectors. There are subtle differences between the behaviour of spinors and vectors under coordinate rotations. Rotating a spinor (spin one-half particle) by 360° does not bring it back to the same quantum state, but to the state with the opposite quantum phase; this is detectable, in principle, with interference experiments. To return the particle to its exact original state, one needs a 720° rotation [19]. A spin-zero particle, on the other hand, can only have a single quantum state, even after torque is applied and rotation of 180° will bring back a spin-2 particle to the same quantum state and so on.

Quantum mechanics says that if a particle is turned around by 360° , its statefunction changes by a phase of either $+1$ or -1 and is related to its spin. Also the symmetry (or changes of phase) of the quantum state of an ensemble of identical particles under interchange of two particles determines the 'statistics' of the assembly, their joint statefunction changes phase either by $+1$ or -1 . Some particles change phase by $+1$ when turned through 360° or two of them in the ensemble swap places. Others change phase by -1 when you rotate one by 360° or switch two of them. According to the 'spin-statistics theorem' these are not independent choices. We always get the same phase in both cases!

Particles with half-integer spin are described by antisymmetric quantum states and not allowed to share single quantum states, a restriction known as the Pauli exclusion principle. These are called

⁷The Klein-Gordon (K-G) equation, with appropriate interpretation, is a correct relativistic equation for spin-0 particles. W. Pauli and V. F. Weisskopf [18] have shown that the the apparent 'problems' with K-G equation were actually manifestation of the new physics and the equation was later revived (in Quantum field theory) to describe elementary spin-0 bosons.

fermions and obey Fermi-Dirac statistics. They include electrons, neutrinos, quarks, and composites like protons, neutrons etc.. Particles with integer spin, on the other hand, occupy symmetric states in which a single quantum state can be shared by any number of such particles. They obey Bose-Einstein statistics, and are known as bosons. Bosons include photons, the W and Z bosons, and gluons. In 1940, Pauli [20] proved the Spin-statistics theorem which relies on both quantum mechanics and the theory of special relativity. Branding this hardly accessible proof as ‘intimidating’ and ‘infamous’, I. Duck and E. C. G. Sudarshan [21] have offered an ‘understandable and elementary proof’ and clarify flaws in some frequently discussed intuitive explanations. However, this theorem occupies a very crucial role in the development of the theory of particle physics.

In 1929, Weyl modified his earlier (classical) idea of 1918 in the language of quantum mechanics by incorporating the additional change of phase to the wavefunction of a system due to a local gauge transformation. Similar proposals also came independently from V. Fock and F. London during the same period. This reinterpretation paves the way for the emergence of ‘gauge theory’ as one of the basic idea of all the present day quantum field theories (QFT) starting with Feynman’s ‘the jewel of physics’ quantum electrodynamics (QED)⁸. Under local gauge transformation, the field energy of the transformed system is changed and in QFT the symmetry is restored by offsetting this change with the addition of a compensating field called the ‘gauge field’. In QED, the gauge field is the four vector potential field for the photon and the problem is to find a suitable transformation to have the standard electromagnetic coupling with the electron/positron field. The gauge invariance here requires that the photon be massless.

In 1954, attempting to resolve some of the great confusions in elementary particle physics, C. N. Yang and R. Mills [22] introduced nonabelian (group having noncommuting elements) gauge theories as models to understand the strong interaction holding together nucleons in atomic nuclei (R. Shaw, working under A. Salam, independently introduced the same notion in his doctoral thesis). Generalizing the gauge invariance of electromagnetism, they have constructed a theory based on the action of the nonabelian $SU(2)$ symmetry group (special unitary group of degree 2) on the isospin doublet (similar to the action of the $U(1)$ group on the spinor fields of QED) of protons and neutrons⁹. But this attempt fails as the (strong) isospin is a global and not a gauge symmetry.

On the other hand, with the ever growing number of ‘fundamental’ subatomic particles appearing in experiments during early 1960s, physicists considered the possibility that hadrons (protons, neutrons and mesons) may be composed of smaller units of matter¹⁰. In 1961 M. Gell-Mann and Y. Ne’eman [23] proposed a particle classification scheme called the Eightfold Way, based on the mathematical symmetry group $SU(3)$, that described strongly interacting particles in terms of fewer, tiny building blocks christened as quarks (a fanciful name taken from James Joyce’s novel ‘Finnegans Wake’ by Gell-Mann). The quark model advanced by Gell-Mann in 1964 provided a simple picture in which all mesons are shown as consisting of a quark-antiquark pair and all baryons as composed of three quarks. At the same time, G. Zweig [24] independently developed a similar theory

⁸QED, is among the most stringently tested theories in Physics. Precision tests of QED consist of measurements of the electromagnetic fine structure constant in different physical systems and performed in low-energy atomic physics experiments, high-energy collider experiments and condensed matter systems which confirm QED as one of the most accurate physical theories constructed so far. It may be also mentioned that QED provides an accurate prediction of electron g-factor.

⁹Nucleons (p, n), pi-mesons and the Δ baryons are three examples of groups of similar mass particles differing in charge by one unit. The charge Q in each case can be considered as due to the orientation of an ‘isospin’ $\mathbf{I}$ (which has nothing to do with the real spin of the particle, but obeys the same addition rules as the quantum mechanical spin) in some hypothetical space, such that Q depends on the third component I_3 of $\mathbf{I}$. Thus the nucleons belong to an isospin doublet (with $I_3 = 1/2, -1/2$), the pions form an isospin triplet (with $I_3 = 1, 0, -1$) and the Δ forms a quadruplet (with $I_3 = 3/2, 1/2, -1/2, -3/2$). The electric charge is to be written as $Q = e(\frac{1}{2}B + I_3)$, where B is the baryon number which is 1 for baryons and 0 for mesons.

¹⁰Both the nonzero spin magnetic moment of electrically neutral neutron and the anomalous g-factor of proton may be cited as the early indications that they are not fundamental but composite particles.

and called his fundamental particles ‘aces’ (R. Feynman also came up with the name ‘parton’ for the parts of hadrons). Initially, Gell-Mann postulated the existence of three types (‘flavours’) of quarks¹¹, now designated as ‘up’ (u), ‘down’ (d), and ‘strange’ (s). Quarks are fermions and each carries a fractional electric charge ($+2e/3$ or $-e/3$). The up and down quarks are thought to make up protons and neutrons and are thus the ones observed in ordinary matter. Strange quarks occur as components of K mesons and various other extremely short-lived subatomic particles that were first observed in cosmic rays but that play no part in ordinary matter.

With the introduction of the concept of symmetry breaking, Yang-Mills’ theory of gauge field was successfully revived in the field theoretic formulation of *weak interaction*. It is well known that radioactive beta decay cannot be explained in terms of strong nuclear or electromagnetic interactions (which explains radioactive alpha and gamma decays respectively) and required the introduction of an additional rare type of fundamental interaction – called the weak interaction, initially described by Fermi’s theory of a contact four-fermion interaction. But, in the mid-1950s T.-D. Lee and C. N. Yang [25] suggested that the weak interaction might violate the law of parity conservation (laws of nature remain the same under the reversal of all spatial axes), which is respected by all the other known fundamental interactions. C. S. Wu and collaborators [26] in 1957 discovered that the weak interaction in fact maximally violates parity, both confirming Yang and Lee’s prediction and signalling the necessity of a revision of Fermi’s theory. Soon after, R. Marshak and E. C. G. Sudarshan [27] (and somewhat later, R. Feynman and M. Gell-Mann [28] also) proposed a V-AV (vector minus axial vector) Lagrangian for the weak interaction, allowing it to act only on left-handed particles and right-handed antiparticles¹². Since the mirror reflection of a left-handed particle is right-handed, this explains the maximal violation of parity. However, this theory allowed the combined symmetry (CP) of charge conjugation and parity operations to be conserved. Physicists were again surprised in 1964 when clear evidence of CP symmetry violation in neutral kaon decay was reported [29]. Though the magnitude of CP violation is very small compared to parity violation in weak interaction, at present it is the only prospective candidate in our hand for the explanation of matter-antimatter asymmetry in the universe.

The identification of weak isospin¹³ as the gauge symmetry of the weak interaction, which connects quark and lepton doublets of left-handed particles in all generations, paves the way for a gauge theory of weak interaction. It involves all fundamental particles except gluons and photons and to be precise, it affects all left-handed leptons and quarks. The weak interaction is unique in a number of respects:

- (1) It is the only interaction capable of changing flavour of elementary particles.
- (2) It is the only interaction which violates parity symmetry – it exclusively acts on left-handed particles (and right-handed antiparticles). It is also the only one which violates CP-symmetry.
- (3) It is mediated by massive gauge bosons $W^\pm$ and Z .
- (4) It is the most important interaction affecting neutrinos.

Due to the change of flavour by weak interaction, only the lightest quarks and leptons are included in the stable matter – all heavier ones decay to the lighter particles. So, despite their slow rate

¹¹Evidence for the existence of quarks comes from deep inelastic scattering: firing electrons at nuclei to determine the distribution of charge within nucleons (which are baryons). If the charge is uniform, the electric field around the proton should be uniform and the electron should scatter elastically. Low-energy electrons do scatter in this way, but above a particular energy, the protons deflect some electrons through large angles. The recoiling electron has much less energy and a jet of particles is emitted. This inelastic scattering suggests that the charge in the proton is not uniform but split among smaller charged particles: the quarks.

¹²Right or left handedness implies the projection of the spin direction onto the momentum vector of the particle being positive or negative respectively.

¹³Weak isospin is a quantum number relating to the weak interaction and parallels the idea of isospin of the strong interaction. Particles not involved in the weak interactions have a value of 0. Other elementary particles have weak isospin values of $-1/2$ or $1/2$. particles not involved in the weak interactions have a value of 0. Other elementary particles have weak isospin values of $-1/2$ or $1/2$.

and short range, *weak interactions play a crucial role in the make-up of the world we observe*. N. Cabibbo [30] first explained the nonconservation of quark flavour in weak interaction by claiming that the eigenstates are those of weak isospin, different from the eigenstates of strong interaction. He formulates a relationship between the two eigenstates (strong and weak) of the first two quark generations. Based on this formalism M. Kobayashi and T. Maskawa [31] discovered the origin of symmetry breaking which predicts the existence of at least three generations of quarks and the relationship is expressed by a 3×3 square matrix known as CKM matrix which explains the CP violation. All these theoretical predictions are vindicated by later experimental findings.

The electro-weak unification by A. Salam and S. Weinberg next reveals that at low energies, the gauge symmetry is spontaneously broken down to the U(1) symmetry of electromagnetism. (The strength of the weak interaction can be characterized by the ‘weak charges’, distinct from the electric charges, of the interacting quarks and leptons. The two charges are comparable which provides an indication of unification. However, we are not going to be involved here into the discussion of ‘weak charges’). Three of the four gauge bosons that mediate the interactions ($W^\pm$ and Z) acquire mass by the so-called Higgs mechanism¹⁴ while the photon (γ) remains massless. This theory has made a number of correct predictions, including the prediction of the Z and W bosons before their discovery. Due to the nonzero rest mass, the interaction-mediating particles decay quickly and thereby make the interactions short-ranged – even shorter than the range of strong nuclear interaction.

Finally the quantum field theory of the strong interaction (quantum chromodynamics - QCD) is developed with the appreciation of colour charge as the source of the strong interaction between quarks and the demonstration of the renormalizability of nonabelian gauge theory by G. ’t Hooft in 1971 [35]. According to QCD, the strong interactions involve the exchange and/or production of massless gauge boson gluons. The gauge theories became even more attractive when it was realized that nonabelian version reproduced important features of strong nuclear interaction called asymptotic freedom and confinement. In 1973, D. J. Gross and F. Wilczek [36] and also H. D. Politzer [37] independently arrived at the important result that though QCD is a strongly coupled theory, at very small distances, which may be probed by scattering at very high energies, the constituents of hadrons (quarks and gluons) behave almost as free particles¹⁵. On the flip side, at larger distances the coupling constant becomes enormously high – a phenomenon known as colour confinement that renders the strong interaction extremely short-ranged. Also as a consequence, the colour charged objects (such as quarks or gluons) are prevented from being isolated.

3. The Standard Model – on the Fast Track of High Energy Physics

The universe is believed to consist of 4.6% visible matter/energy, 23.3% dark matter and 72.1% dark energy (results from WMAP) and the Standard Model of particle physics describes only the visible matter/energy and their interactions in terms of 12 fermions (6 leptons and 6 quarks), the corresponding antiparticles, and three (electromagnetic, weak nuclear and the strong nuclear interactions) out of the four known fundamental interactions. It also cannot account for the weakest interaction – gravitation! Despite some other weaknesses too, it is the established paradigm of particle physics and explains almost all experimental data. The SM has requisitioned two major extant theories

¹⁴Three independent seminal papers [32, 33, 34] culminated to present a consistent theoretical framework which explains how massless gauge bosons of (nuclear) weak interaction acquire mass. It was Higgs insight to predict one or more new massive scalar bosons finally christened as ‘Higgs bosons’.

¹⁵In 1969 Iosif Khriplovich correctly calculated the coupling constant in the SU(2) gauge theory without emphasizing the implication of the result and G. ’t Hooft in 1972 also noted the effect but missed the physical significance and did not publish [38].

– gauge theories of quantum electroweak and quantum chromodynamics – into an internally consistent theory describing the three interactions between all experimentally observed particles. The three fundamental interactions are mediated by the spin-1 gauge bosons – γ , $W^\pm$, Z and gluons (see Table - 1). The various elementary particles of SM are described by different equations (limit of the corresponding quantum fields) according to their spins. The spin-0 boson is described by the Klein-Gordon equation, the spin- $\frac{1}{2}$ fermions by Dirac equation, the spin-1 massive vector bosons by Proca equation [39] whereas, QED provides an accurate description of spin-1 massless photons. Recently, K. Pachucki [40] has developed a technique by identifying the mathematical terms that are necessary to construct the right interaction Hamiltonian to provide the description for relativistic particles of arbitrary spin.

Table 1. Four fundamental interactions of nature

Interactions	Acts on	Relative strength	Range in m	Mediating particles & Mass in GeV
Strong nuclear	Colour charge	1	10^{-15}	gluon (0.14)?
Electromagnetic	Electric charge	$1/137$	∞	photon (0)
Weak nuclear	Flavour	10^{-6}	10^{-16}	$W^\pm$ & Z bosons (80 - 90)
Gravitation	Mass & energy	10^{-39}	∞	graviton? (0)?

S. Glashow [41] took the first step towards the Standard Model in 1961 by inventing a way to combine the electromagnetic and weak interactions. In 1967, S. Weinberg [42] and A. Salam [43] incorporated the Higgs mechanism¹⁶ into Glashow's electroweak theory giving it the present form. After the discovery of neutral weak currents caused by Z boson exchange (at CERN in 1970's), the electroweak theory became widely accepted.

According to the spin-statistics theorem, all the fermions of SM obey Pauli exclusion principle. Also, each fermion has a corresponding antiparticle having the same mass, opposite electric charge (for charged particles) and opposite handedness. Shortly after the existence of quarks was first proposed in 1964, O. W. Greenberg [44] introduced the notion of an additional quantum degree of freedom (later to be known as colour charge) to explain how quarks could coexist inside some hadrons (e.g. spin- $\frac{3}{2}$ Delta baryons) in otherwise identical quantum states without violating the Pauli exclusion principle¹⁷. The concept turned out to be very useful and most problems with quarks were resolved by the introduction of the concept of colour charge. The three kinds of colour charge (as opposed to only one kind of electric charge) combine to produce zero net colour charge and are

¹⁶In order to endow particles with mass, a background field is invented which becomes locally distorted whenever a particle moves through it. The distortion - in the form of a clustering of the field around the particle – generates the particle's mass. The idea comes directly from the Physics of Solids. When an electron moves through a crystal lattice it attracts the atoms at the lattice sites and the effective mass of the electron becomes much greater (as much as 40 times) than the mass of a free electron. The postulated Higgs field in the vacuum is a sort of hypothetical lattice which fills our Universe.

A crystal lattice can also carry waves of clustering without needing a particle to move through and attract the atoms. These waves/disturbances behave as phonon-like quasi-particles. The Higgs boson is predicted to be just such a clustering in the Higgs field. There could be a Higgs mechanism, and a Higgs field all throughout, with or without there being a Higgs boson. High energy physics is searching for the answer with the next generation of colliders.

¹⁷Perhaps the first remark that quarks should possess an additional quantum number was made in a short footnote of a preprint (submitted in early 1965) by B. Struminsky (a Ph.D. student of N. Bogolyubov) on the composition of Ω^- hyperon (three strange quarks with parallel spins). A detailed discussion was presented subsequently by N. Bogolyubov, B. Struminsky and A. Tavcheldidze.

referred to as red, green and blue by loose analogy to the three primary colours (red, green and blue) perceived by humans.

Colour charge has two aspects: (a) as a quantum number that labels states of quarks in a hadron like composite and (b) as the source of the strong field acting between colour charges of quarks (analogous to aspects of electric charges as the source of electromagnetic fields acting between electrically charged particles). M.-Y. Han and Y. Nambu [45] introduced the aspect of colour charge as the source of the strong interaction between quarks. With the stipulation of conservation of colour charge, a quantum field theory of colour-charged fermions (the quarks) becomes an automatic extension of the Yang-Mills' nonabelian gauge theory. Since the quantum field theory of electric charge is dubbed electrodynamics, the Greek word 'chroma' has prompted the appellation 'chromodynamics' for the theory of colour charge.

In the Standard Model the fundamental fermions form two groups, leptons and quarks, according to how they interact or by what charges they carry. Pair (doublet) taken from each group form a generation. In the standard electroweak gauge theory, the discrete C-, P- and T (time reversal) symmetries are violated (the Lagrangian is not invariant under any of these operations) and it requires that quarks and leptons come in three generations.

Three of the six leptons have electrical charge equal to $-e$ and are divided into three lepton families, each accompanying a chargeless neutrino in its family. Leptons are point-like particles without any internal structure. The best known lepton is the electron (e^-) and the other two charged leptons are called muon (μ^-) and tauon (τ^-), which are more massive. The other leptons are the three neutrinos (ν), having no electrical charge and very little mass. They are really very hard to be detected.

Each quark has one of the fractional electric charges, either $+2/3$ or $-1/3$ in unit of e . But free-standing quarks have never been observed – the theoretical reason for this fact is confinement and asymptotic freedom. Hadrons (baryons and mesons), which interact via strong interaction, are composite particles made-up of quarks. Baryons (e.g. protons, neutrons etc.) are fermionic hadrons and made up of three quarks (qqq) while mesons, the bosonic hadrons contain one quark (q) and one antiquark (q-). *The quarks are then a crazy lot – always huddle together to feel free within the confines of baryons and can peacefully coexist with the antiquarks in mesons!* Although individual quarks have fractional electrical charges, they combine such that hadrons have a net integral electric charge. Another property of hadrons is that they have no net colour charge even though the quarks themselves carry colour charge.

In the idealized version of the Standard Model, the quarks are considered as massless particles and the acquisition of mass by elementary fermions is explained in the realistic (standard) model by invoking the 'Higgs mechanism'. The sum of the masses of the independent quarks in a hadron is far less than the mass of the hadron they spawn. The mass is mostly there because of the kinetic and potential energy of the quarks, converted to mass according to

$$m = \frac{E}{c^2}.$$

The classification of the fermion ingredients of the theory into three generations is in accordance with the mass spectrum of the fermion doublets (u, d), (e, ν_e), ... etc. – one lepton flavoured and one quark flavoured (in three colours) doublets are in each generation. A fermion with a given charge in a given generation is lighter than the corresponding fermion in a higher generation. With $m_u < m_c < m_t$, u , c and t are placed in the first, second and third generation respectively and correspondingly e , μ and τ (as $m_e < m_\mu < m_\tau$). For neutrinos, the one which couples most strongly to the electron is called ν_e and is placed in the first generation, and so on (Table - 2).

The fermion assignment of the Standard Model is free from gauge anomalies, as each generation is anomaly-free with respect to the gauge bosons. In the electroweak gauge theory, higher-order quantum corrections to the interaction of three gauge bosons by exchanging a fermion in a loop may lead to infinite masses for the gauge bosons (W, Z, and photon) and are called 'anomalies' [46].

Table 2. Elementary fermions of SM

First generation	Second generation	Third generation	Charge (in unit of $ e $)
Up quark (u)	Charm quark (c)	Top quark (t)	$+2/3$
Down quark (d)	Strange quark (s)	Bottom quark (b)	$-1/3$
Electron e^-	Muon μ^-	Tauon τ^-	-1
Electron neutrino ν_e	Muon neutrino ν_μ	Tauon neutrino ν_τ	0

In the case of photon which interacts with charged particles with a strength proportional to their charges, the anomaly is removed if the sum of all the charges of the quarks and leptons within each generation add up to zero. There are three colours associated with each of the quarks, and writing the component charges as Q_i s, the total charge of the fermions in the first generation is:

$$\begin{aligned}
 Q &= 3(Q_u + Q_d) + (Q_e + Q_{\nu_e}) \\
 &= 3\left(\frac{2}{3} - \frac{1}{3}\right) + (-1 + 0) = 0.
 \end{aligned} \tag{6}$$

Similarly, the anomaly cancelling criterion (charge relationship) is satisfied in each generation of fundamental fermions.

In the following section, we discuss the present status of our understanding regarding the origin of the three basic attributes of matter.

4.1. . . . and let there be MASS!

In his celebrated 1905 paper entitled “Does the inertia of a body depend on its energy content?”, A. Einstein suggests the possibility of explaining mass in terms of energy. In modern particle accelerators, the equation $m = E/c^2$ comes to life and mass has been indeed created from energy in the Large Electron-Positron (LEP) collider. On the other hand, after gigantic and totally opaque calculations involving the equations of QCD, the super computers are now able to ‘construct’ massive particles from the nuclear building blocks – quarks and gluons! These calculations actually account for about 90% of the nucleonic (proton and neutron) mass, emerging from the idealized theory whose ingredients are massless u and d quarks and massless gluons. The rest 10% mass is accounted for in the realistic theory using nonzero mass for u and d quarks and the four additional flavours of quarks (which do not directly appear in ordinary matter but have some effect through participation as virtual particles).

In a stimulating article, F. Wilczek [13] has elucidated the understanding hidden behind the intricate computer calculations for the ‘origin of most of the mass of ordinary matter’. The concept of asymptotic freedom predicts enhanced coupling of lone colour charge to soft radiation and finally inhibits a colour charge in isolation. The enhancement leads a runaway growth to a big colour thundercloud. But all these involve infinite energy which, even our Mother Nature can’t afford! This is the theoretical explanation why lone-standing quarks have never been observed – they always huddle together.

We are already told about the two ways of forming a stable composite by quarks: one by a quark-antiquark pair and another by huddling together three quarks, one of each colour, to cancel the colour charge. But then, microparticles do resist precise location and quantum mechanics allows a number of compromise stable solutions (wave patterns) with quark and antiquark (or three quarks) in close proximity. Each wave pattern corresponds to a different kind of particle (proton/neutron/ .. etc.) and a characteristic energy which appears as the particle mass according to the ‘world’s most

famous equation'. We should note here that the understanding of the origin of mass in standard model still remains incomplete without invoking 'Higgs mechanism' for the acquisition of mass by elementary fermions, like electron and quarks.

4.2. Charged up – how?

In the standard model, the elementary particles are endowed with charges and specially quarks have two types of charges – 'electric' and 'colour'. While quantum electrodynamics (QED) deals with only one type of charge for the leptons, describing the response of photons to the 'electric charge', quantum chromodynamics (QCD) describes the quark-gluon interaction in terms of their 'colour charges'.

When the 'electric charge' is contrasted with the subnuclear 'colour charge', their nature can be understood with greater clarity from the symmetry argument. The symmetry of colour charge is described by the group SU(3), while the symmetry of electrical charge is described by the group U(1). Consequently, there are three kinds of colour charge and only one kind of electric charge. The term 'charge' by itself is synonymous with plain old 'electric charge', but the other 'charge' of a quark is always referred to as 'colour charge'.

All known elementary particles that can be isolated have an electric charge ne where e is the magnitude of the electron's charge and n is 0 or an integer. The search for elementary particles with fractional electric charge in bulk matter using advanced forms of the Millikan oil drop experiment so far yield negative results. However, two groups of physicists [47, 48] have reported in 1997 the experimental evidence of electric current being carried by fractionally charged quasiparticles. The results agree quite well with the theory of fractional quantum Hall effect, formulated by R. Laughlin in 1982. According to Laughlin [49], electrons in strong magnetic fields form an exotic collective state and quasiparticles or quanta representing the collective state carry fractional charges.

The questions remain, why charge is quantized and what gives the quark/lepton its charge? Different particles have all sorts of different ratios of masses devoid of any pattern and those ratios aren't even whole numbers. Yet those same particles have very specific charges: either plus or minus one electron charge or zero. Something special about the electric charge that makes it come in these fixed-size lumps, or charge quanta. We can advance here only a consistency argument from the requirement of cancellation of anomalies in the gauge theory.

The requirement of anomaly cancellation in the electroweak gauge theory demands that the charges on the quarks and leptons must be related (eqn. 6). But it cannot predict that the proton and the electron have equal and opposite charge¹⁸. However, the spontaneous decay of a free neutron into an electron, an electron antineutrino and a proton tells something in this regard. According to the quark model¹⁹, the charge on a proton is that of two up quarks and one down quark, $(2 \times 2/3 - 1/3) = 1$, and one up quark and two down quarks make neutron electrically neutral. What happens at the quark level is that a down quark emits a W^- boson, turning into an up quark in the process. The W^- boson then couples to the electron and electron antineutrino. This decay indicates that the difference between an up quark's charge and a down quark's charge is the charge on an electron. This model and the above anomaly relation is thus consistent with the fact that proton and electron have opposite charges.

We have thus arguments explaining the connection between the existence of electric charge and the structure of the theory describing the elementary particles, but this still doesn't answer the questions why and how? – the origin of charge remains one of the mysteries of physics. In this respect the conjecture, that the charge-current density of electromagnetism may have its roots in the

¹⁸More speculative 'Grand Unified Theories' (models unifying the strong and electroweak forces) propose that the separate symmetries of the strong and electroweak interactions are only part of a much larger class of symmetries which have not yet been observed. Some of these models predict together with some weird stuff like leptoquarks etc. the charge equality of the electron and proton (with the minus sign).

¹⁹An extensive review of the quark model is given by S. Gasciorowicz and J. L. Rosner [50].

differential geometry and topology of curvature and torsion, may provide some clue to the origin of charge.

4.3. On the Spinning track:

Leptons and quarks are elementary particles that cannot be divided into smaller units. The spin possessed by these particles cannot be explained by postulating that they are made up of even smaller particles rotating about a common center of mass (having some internal structure). According to the standard model an elementary particle is a true point particle and discarding the ‘self rotating’ particle model, spin is considered as an intrinsic physical property.

On the other hand, it is not always easy to deduce the spin of a composite particle, *ab initio*, from the spins of the constituent particles. For example, even if we know that the proton is a spin- $\frac{1}{2}$ particle, it is not clear how this spin is distributed among the three internal valence quarks, the surrounding sea quarks and gluons and the issue is an active area of research [51]. However, the naive quark model with three valence quarks provides reasonably good estimate of the anomalous magnetic moments of proton and neutron (and more accurately the ratio of the two) considering the uncertainties in the quark masses [52].

The lack of a proper understanding of the origin (physical mechanism) of spin is criticized by H. C. Ohanian as the acceptance of “merely an extra nonorbital angular momentum of unknown etiology” [53]. The usual claim that spin arises in a natural way from Dirac’s equation is also an exaggeration as Ohanian rightly points out “without prior knowledge of the spin of the electron, we cannot know that Dirac’s equation is applicable”. On the basis of an old calculation by F. J. Balinfante, he argued that the (electron) spin may be regarded as an angular momentum generated due to a circulating flow of energy, or a momentum density, in the electron wave field. Similarly, starting from the Gordon decomposition [19] of Pauli current density (Dirac current density in the nonrelativistic limit), E. Recami and G. Salesi [54] have expressed the Bohm’s *quantum potential*²⁰ of a spin- $\frac{1}{2}$ particle as the kinetic energy for the internal ‘zitterbewegung’ (German word for ‘trembling motion’) and identify it as a consequence of the spin motion of the system. Subsequently S. Esposito [56] has generalized the result for arbitrary spin system and reiterated the claim that the “quantum behaviour of microsystems is a direct consequence of the fundamental existence of spin”. It is also argued that the internal ‘zitterbewegung’ disappears in the limit of zero quantum potential - the supposed classical limit in the causal (de Broglie-Bohm) interpretation. However, it should be emphasized here that we cannot proceed to the limit for an elementary particle. Even in the case of a many particle system, macroscopic spin arises (as in the Einstein-de Haas effect and the magnetization found in permanent magnets) in the limit of large number of interacting particles. For an assembly of large number of interacting microscopic particles, entangled macroscopic spin states showing strong nonclassical features also exist [57]. The commonly referred quantum mechanical macrostates are essentially manifestation of collective behaviour of internal modes of macroscopic systems. Single statefunction description is possible for coherent many-body system representing a quantum macrostate which interacts as a single entity, exhibiting collective excitations and the constituents of the system all move coherently in phase giving rise to strong nonclassical features [58].

With the use of spinor fields for the description of an elementary particle, the spin of the particle is introduced naturally in relativistic quantum equation of motion. According to Einstein and Mayer [59], the irreducible representation of the symmetry group of special relativity theory imply that the two-component spinor variable is the most primitive way to express the laws of nature. This led M. Sachs to conclude [60] that, the spin is “clearly a consequence of the theory of relativity”.

²⁰The extra nonclassical term containing $\hbar$ in the quantum mechanical Hamilton-Jacobi equation [55] which together with the (quantum mechanical) continuity equation forms a pair of coupled equations that follows directly from the Schrödinger equation [6].

Some physicists even prefer to consider spin as ‘rest angular momentum’ à la the concept of rest mass. But, it appears to be a gross simplification of a complicated notion since the introduction of spin is required by the conservation of angular momentum in the quantum mechanical equation of motion for a charged particle in a central field. Dirac’s equation first provided a coherent theoretical framework in which spin solves the problem of conservation of angular momentum and facilitates construction of the generators of the rotation group of the relevant group theory [53, 61]. On the other hand, J. M. Levy-Leblond’s analysis [62] has clearly demonstrated that it is possible to develop a nonrelativistic theory of electron spin. A fully articulated formulation describing interaction with electromagnetic field, however, requires an association with relativity.

From an analysis of the solution Dirac equation for relativistic electron in free space, Schrödinger in 1930 first predicted a trembling motion (‘Zitterbewegung’) due to the interference between positive and negative energy states. Taking a cue from Schrödinger’s work a fringe theory *The Wave Structure Matter* (WSM) model claims that particles are the appearance (‘schaumkommen’ in the words of Schrödinger) of wave structures in space and the spin of the electron (or any other charged elementary particle) arises as a result of rotation, either clockwise or anticlockwise, of the inward spherical quantum wave (of the electron) at the wave-centre to return as the outward wave (the wave rotation is required to maintain proper phase relations of the wave amplitudes) [63]. Of the two possible combinations of the rotating inward and the outward waves, one represents an electron and the other a positron. The existence of a purely geometrical model/property of space called *spherical rotation* [64], that satisfies Dirac’s equation, permits an object structured in 3D space to spin about any axis without disrupting the continuity of space. This property allows the electron (located at the wave-centre) to retain spherical symmetry while imparting a quantized *spin* along an arbitrary axis as the spherical waves continually move inward and outward. The allowed motions of a particle with spherical symmetry must be represented by a compact simply-connected group. The most elementary of such group is SU(2) which provides all the known properties of electron spin. By introducing the concept of ‘wave structures’ for electrons, WSM model thus claims to provide a physical origin of electron spin in terms of ‘wave rotations’.

However, serious attention and appreciation require some concrete experimental predictions from the model. Also we are not told in this theory, where from the initial torque comes. N. Hamein and E. A. Rauscher [65] have claimed that their solution to the modified Einstein’s field equations, termed as Hamein-Rauscher metric, include ‘the fundamental forces necessary to generate spin’ and provides a possible description for the origin of spin. It is posited that the required torque comes “from a change in density (or gradient) in the geometry of spacetime” in their theory. But the introduction of concept like Newtonian Coriolis force, which is clearly not relativistically covariant, raises serious question and again there seem to be no experimental predictions from the theory.

5. Lacking an epilogue

After all this recapitulation, it appears that we have an inkling of the origin of particle mass, but in the cases of spin and charge we have framed the questions only and are far from convincing answers regarding their physical origins. The structure of the basic equations are intimately connected with the existence of electric charge and spin in a consistent way, but their origin are not properly explained in the theory. For a better understanding, physicists are now looking beyond the standard model.

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